

Producer Consumer Problem

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int mutex = 1;
```

```
int full = 0;
```

```
int empty = 10, data = 0;
```

```
void producer()
```

```
{
```

```
    --mutex;
```

```
    ++full;
```

```
    --empty;
```

```
    data++;
```

```
    printf("\nProducer produces item number: %d\n", data);
```

```
    ++mutex;
```

```
}
```

```
void consumer()
```

```
{
```

```
    --mutex;
```

```
    --full;
```

```
    ++empty;
```

```
printf("\nConsumer consumes item number: %d.\n", data);
data--;
++mutex;
}

int main()
{
    int n, i;
    printf("\n1. Enter 1 for Producer"
           "\n2. Enter 2 for Consumer"
           "\n3. Enter 3 to Exit");

    for (i = 1; i > 0; i++)
    {
        printf("\nEnter your choice: ");
        scanf("%d", &n);
        switch (n)
        {
            case 1:
                if ((mutex == 1) && (empty != 0))
                {
                    producer();
                }

                else
                {
                    printf("The Buffer is full. New data cannot be
produced!");
                }
                break;
        }
    }
}
```

case 2:

```
        if ((mutex == 1) && (full != 0))
        {
            consumer();
        }

        else
        {
            printf("The Buffer is empty! New data cannot be
consumed!");
        }
        break;

    case 3:
        exit(0);
        break;
    }
}
}
```

Output:

```
G:\OS\lab5>producer_consumer.exe
```

1. Enter 1 for Producer
2. Enter 2 for Consumer
3. Enter 3 to Exit

Enter your choice: 1

Producer produces item number: 1

Enter your choice: 2

Consumer consumes item number: 1.

Enter your choice: 2

The Buffer is empty! New data cannot be consumed!

Enter your choice: 1